

Model Explainability under Epistemic Uncertainty

① Introduction

You will be shown outputs from different decision-support systems that estimate the risk of a human health condition. Your task is to interpret these outputs. There are no right or wrong answers. We are interested in how you understand and use the information provided. Please answer quickly and intuitively.

Different systems may present results in different ways:

- A single number (e.g., 0.70) usually represents an estimated risk (the probability of the condition).
- An interval (e.g., 0.50–0.80) indicates that the true risk may lie within that range.
- Some systems may additionally indicate uncertainty due to limited or conflicting information.

Risk is a value between 0 and 1, with smaller values indicating a lower risk of the condition and higher values with a higher risk.

You do not need technical knowledge to answer the questions --- just use your intuition!

Scenario 1

System I estimates that the risk that a person has a certain condition is 0.72.

Aussage	strongly disagree	disagree	neutral	agree	strongly agree
I understand what this prediction means	0%	0%	19%	45%	35%
This output clearly communicates uncertainty	10%	26%	19%	32%	13%
I feel confident that I correctly interpreted this output	0%	13%	29%	35%	23%
This output is easy to understand and interpret	6%	6%	16%	42%	29%
This output helps me make a decision	3%	10%	23%	45%	19%

System II estimates that the risk of a condition supported by the available evidence is 0.70.

Aussage	strongly disagree	disagree	neutral	agree	strongly agree
I understand what this prediction means	3%	0%	19%	55%	23%
This output clearly communicates uncertainty	3%	19%	26%	26%	26%
I feel confident that I correctly interpreted this output	0%	6%	29%	42%	23%
This output is easy to understand and interpret	0%	6%	16%	65%	13%
This output helps me make a decision	0%	10%	23%	48%	19%

System III estimates that the risk of the condition, as supported by the available evidence, lies between 0.66 and 0.74. This range reflects uncertainty due to incomplete information about one of the patient's risk factors.

Aussage	strongly disagree	disagree	neutral	agree	strongly agree
I understand what this prediction means	0%	0%	13%	65%	23%
This output clearly communicates uncertainty	0%	10%	23%	35%	32%
I feel confident that I correctly interpreted this output	0%	3%	32%	52%	13%
This output is easy to understand and interpret	0%	3%	13%	74%	10%
This output helps me make a decision	0%	3%	13%	58%	26%

System IV estimates that the risk of the condition, as supported by the available evidence, lies between 0.66 and 0.74. This range reflects uncertainty in the model.

Aussage	strongly disagree	disagree	neutral	agree	strongly agree
I understand what this prediction means	0%	0%	29%	52%	19%
This output clearly communicates uncertainty	0%	3%	16%	39%	42%
I feel confident that I correctly interpreted this output	0%	6%	32%	42%	19%
This output is easy to understand and interpret	0%	6%	26%	65%	3%
This output helps me make a decision	3%	3%	29%	58%	6%

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⁸² System V estimates that the risk of the condition, as supported by the available evidence, lies between a lower bound of 0.64–0.68 and an upper bound of 0.72–0.76. This “double” range reflects uncertainty due to incomplete information about one of the patient’s risk factors, as well as uncertainty in the model itself.

Aussage	strongly disagree	disagree	neutral	agree	strongly agree
I understand what this prediction means	0%	3%	52%	35%	10%
This output clearly communicates uncertainty	0%	10%	16%	35%	39%
I feel confident that I correctly interpreted this output	0%	6%	55%	32%	6%
This output is easy to understand and interpret	6%	23%	39%	23%	10%
This output helps me make a decision	3%	3%	58%	26%	10%

⁸³ System VI estimates the risk of the condition as 0.72, with evidence indicating a conflict of 0.37 and a level of incomplete information of 0.06.

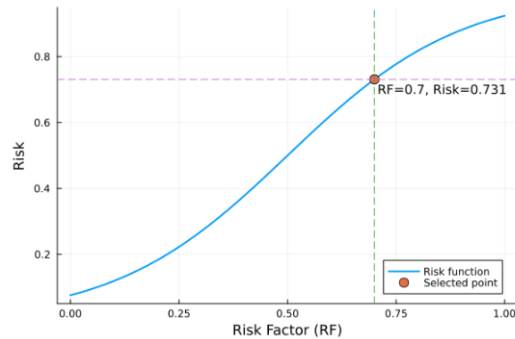
Aussage	strongly disagree	disagree	neutral	agree	strongly agree
I understand what this prediction means	0%	10%	39%	42%	10%
This output clearly communicates uncertainty	0%	3%	19%	65%	13%
I feel confident that I correctly interpreted this output	0%	10%	61%	23%	6%
This output is easy to understand and interpret	0%	52%	19%	23%	6%
This output helps me make a decision	0%	13%	58%	26%	3%

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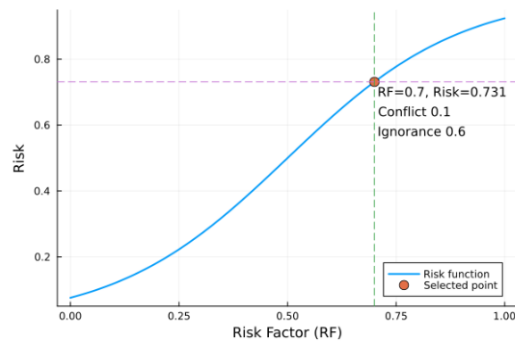
Scenario 2

You are shown how the estimated risk of a condition varies depending on a risk factor. The relationship is uncertain due to incomplete information. Four decision-support systems (A, B, C and D) provide different representations of this risk and its uncertainty (see Figures). Please compare the outputs and answer the questions below.

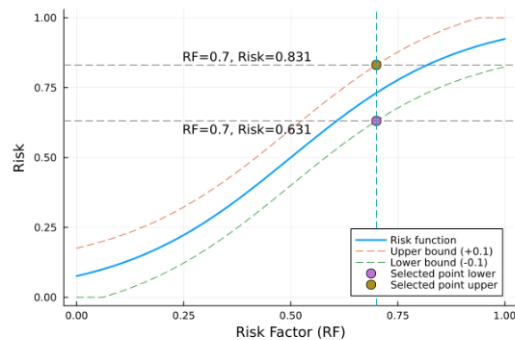
System A



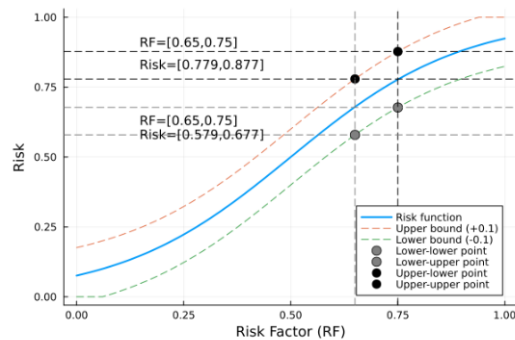
System B



System C



System D



I understand the prediction itself.

Aussage	strongly disagree	disagree	neutral	agree	strongly agree
System A	0%	10%	29%	29%	32%
System B	0%	13%	29%	52%	6%
System C	0%	3%	39%	52%	6%
System D	0%	13%	45%	26%	16%

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■ The visualization helps me understand the prediction and its uncertainty.

Aussage	strongly disagree	disagree	neutral	agree	strongly agree
System A	0%	23%	32%	39%	6%
System B	0%	16%	23%	52%	10%
System C	0%	10%	16%	42%	32%
System D	0%	10%	23%	48%	19%

■ Comparative questions

Aussage	System A	System B	System C	System D	no difference
Which output most appropriately represents the uncertainty in the risk prediction?	16%	6%	16%	55%	6%
Which output would you trust least for making a decision in the presence of uncertainty?	55%	6%	16%	19%	3%
Which output is easiest to understand given that uncertainty is present in the prediction?	19%	19%	32%	23%	6%
Which output helps you best make a decision when the prediction is uncertain?	6%	6%	35%	48%	3%
Which output is easiest to understand in terms of how the prediction was produced and should be interpreted?	45%	10%	6%	29%	10%

■ How well do you understand what this output means for decision-making?

Aussage	very bad	bad	neutral	good	very good
System A	3%	13%	26%	35%	23%
System B	0%	10%	35%	45%	10%
System C	0%	3%	16%	32%	48%
System D	3%	3%	39%	29%	26%

① Thank you!